Software Architecture Design

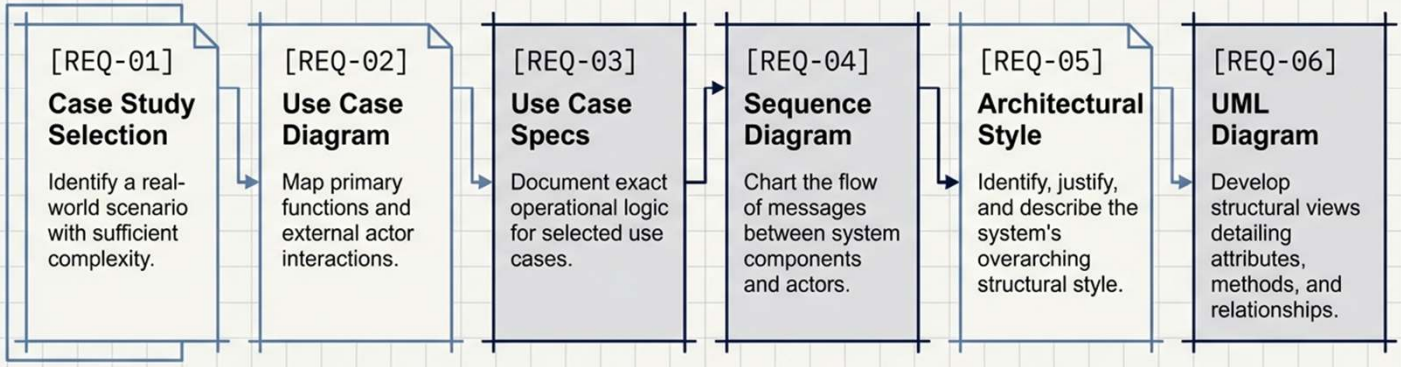
Assignment 1 Playbook

[MISSION-BRIEF]

OBJECTIVE	Design, analyze, and model a software architecture solution.
CONTEXT	Applied to a real-world case study.
CORE CHALLENGE	Apply fundamental software architecture principles and design techniques, bridging the gap between high-level use cases and concrete architectural diagrams.

SYSTEM_STATUS: INITIALIZING

The assignment follows a strict six-phase engineering pipeline



Task distribution separates global architecture from individual module execution

[GLOBAL_ARCHITECTURE]

Group Scope

The team works collaboratively to establish the foundation, map the global use cases, and select the overarching architectural style.

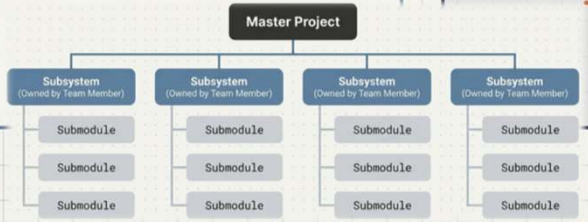
TASK 1 | TASK 2 | TASK 5

[MODULE_EXECUTION]

Individual Scope

Each member is assigned distinct system modules to detail independently.

TASK 3 | TASK 4 | TASK 6



CRITICAL RULE

One module can contain multiple use cases. Individual work must be explicitly labeled in the final report (e.g., 2.1 Sequence for Module 1 - Member 1).

The group establishes the system foundation and overarching architectural style

[REQ-05]

Architectural Style Development

The team must select an architectural style. Crucially, the prompt requires the team to justify this selection by explaining explaining the rationale and proving how it supports both the functional and non-functional requirements of the system.

[REQ-02]

Use Case Diagram

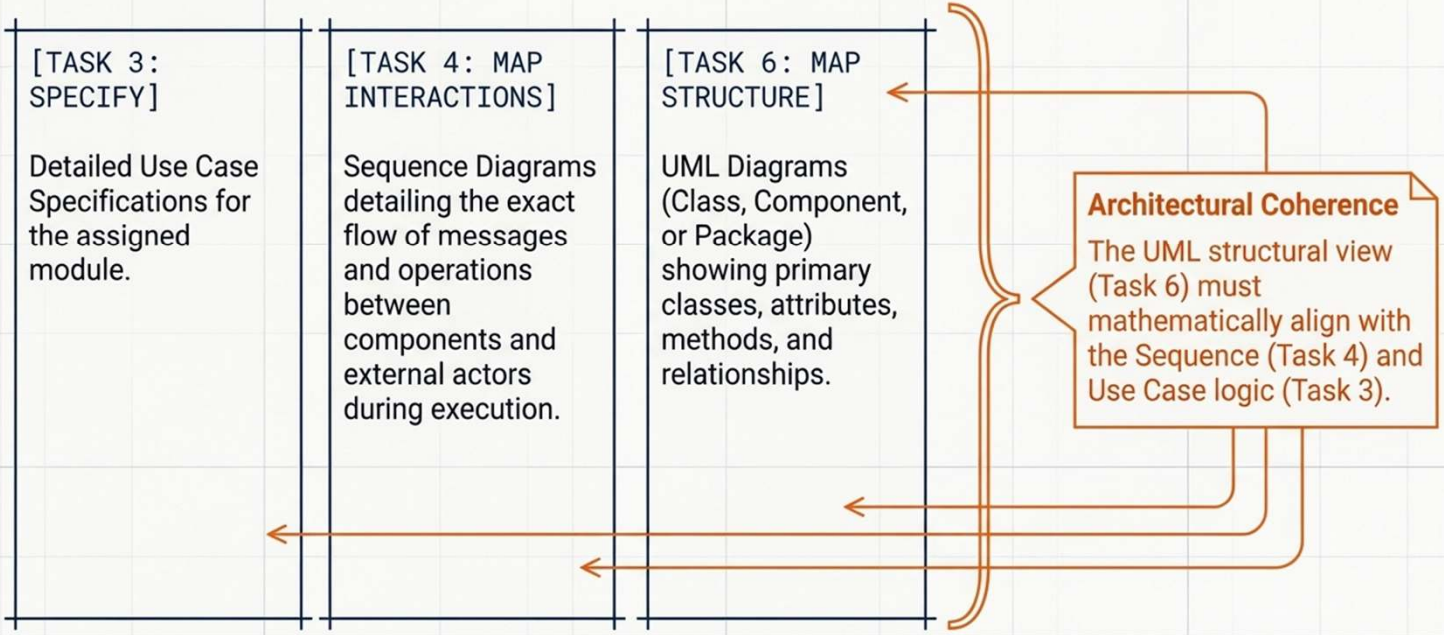
A high-level mapping that locks in the system's primary functionality and its interactions with various external actors.

[REQ-01]

Case Study Selection

Must be a real-world application domain. The chosen case must possess sufficient complexity to warrant deep architectural analysis.

Individual members must design, specify, and align their assigned system modules



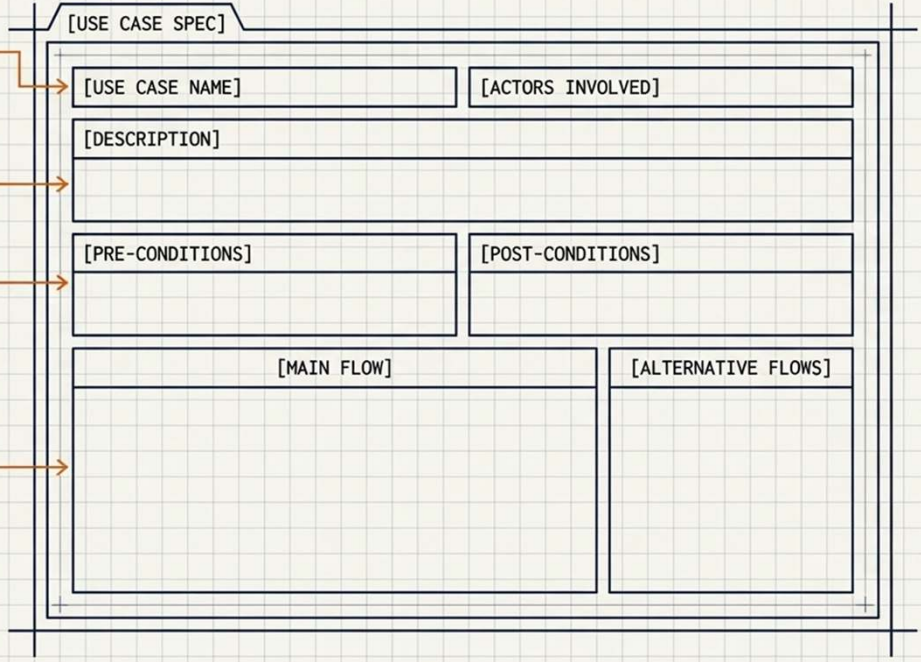
Every module requires a standardized use case specification

[USE CASE NAME] & [ACTORS INVOLVED]: Clearly identify the operation and the participants.

[DESCRIPTION]: A concise summary of the objective.

[CONDITIONS]: Split into Pre-conditions (system state before execution) and Post-conditions (system state after execution).

[FLOW OF EVENTS]: Visually split into a central column for the Main Flow and branching side-paths for Alternative Flows (if applicable).



Final submission requires a single consolidated report by the end of the mid-semester break

LOGISTICS



Format

A single, consolidated report containing all diagrams and supporting documentation.



Deadline

End of the Mid-Semester Break.

[FINAL QA CHECKLIST]

- [ ] Are all diagrams clear, legible, and properly labeled?
- [ ] Are all explanations concise and directly relevant?
- [ ] Is individual work (Tasks 3, 4, 6) explicitly labeled with the assigned member's name and module?
- [ ] Is the global architectural style adequately justified against non-functional requirements?